

ARRANGE TABLES

Bùi Tiến Lên

01/01/2020



KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN

Contents



1. Express: Quantitative Values
2. Separate, Order, and Align: Categorical Regions
3. Spatial Axis Orientation
4. Spatial Layout Density



The Big Picture

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Arrange Tables

① Express Values



② Separate, Order, Align Regions

→ Separate



→ Order



→ Align



→ 1 Key
List



→ 2 Keys
Matrix



→ 3 Keys
Volume



→ Many Keys
Recursive Subdivision



③ Axis Orientation

→ Rectilinear



→ Parallel



→ Radial



④ Layout Density

→ Dense



→ Space-Filling





Why Arrange

- The **arrange** design choice covers all aspects of the use of spatial channels for visual encoding
- It is the most crucial visual encoding choice because the use of space dominates the user's mental model of the dataset.
- The three highest ranked effectiveness channels for quantitative and ordered attributes are all related to spatial position.

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Arrange by Keys and Values

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

- **Key**
 - independent attribute and used as unique index to look up items
 - can be categorical or ordinal
 - simple tables: one key
 - multidimensional tables: multiple keys
- **Value**
 - dependent attribute, value of cell
 - can be all three of the types: categorical, ordinal, or quantitative
- **Classify** arrangements by key count: zero, one, two, many...
- The core design choices for visually encoding tables directly relate to the semantics of the table's attributes: how many keys and how many values does it have?



Express: Quantitative Values



Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

- Using space to express quantitative attributes is a straightforward use of the spatial position channel to visually encode data. The attribute is mapped to spatial position along an axis.
- In the simple case of encoding a single value attribute, each item is encoded with a mark at some position along the axis.
- Additional attributes might also be encoded on the same mark with other nonspatial channels such as color and size.
- In the more complex case, a composite **glyph** object is drawn, with internal structure that arises from multiple marks. Each mark lies within a subregion in the glyph that is visually encoded differently, so the glyph can show multiple attributes at once.



Example: Scatterplots

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

- The idiom of **scatterplots** encodes two quantitative value variables using both the vertical and horizontal spatial position channels, and the mark type is necessarily a point.



Example: Scatterplots (cont.)

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

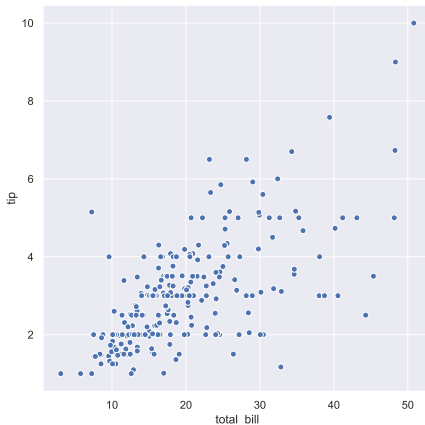
Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

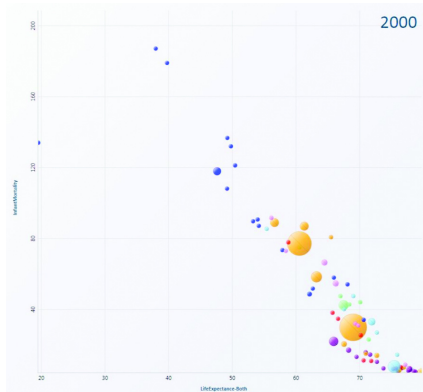
Space-Filling





Example: Scatterplots (cont.)

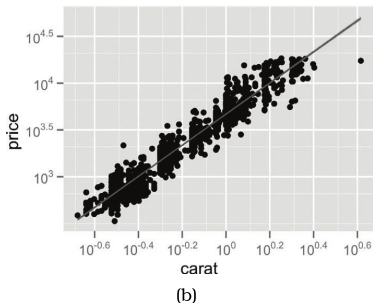
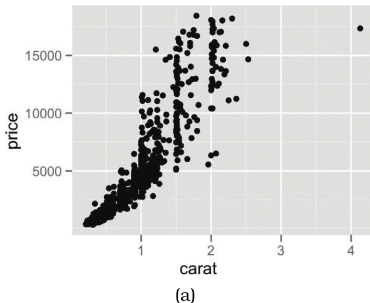
- Each point mark represents a country, with horizontal and vertical spatial position encoding the primary quantitative attributes of life expectancy and infant mortality. The color channel is used for the categorical country attribute and the size channel for quantitative population attribute





Example: Scatterplots (cont.)

- (a) Original diamond price/carat data.
- (b) Derived log-scale attributes are highly positively correlated.





Idiom

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Idiom	Scatterplots
What: Data How: Encode Why: Task Scale	Table: two quantitative value attributes. Express values with horizontal and vertical spatial position and point marks. Find trends, outliers, distribution, correlation; locate clusters. Items: hundreds.



Separate, Order, and Align: Categorical Regions

- List Alignment: One Key
- Matrix Alignment: Two Keys
- Volumetric Grid: Three Keys
- Recursive Subdivision: Multiple Keys



Separate, Order, and Align: Categorical Regions

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

- The use of space to encode categorical attributes is more complex than the simple case of quantitative attributes where the value can be expressed with spatial position.
- Spatial position is an ordered magnitude visual channel, but categorical attributes have unordered identity semantics.
- The semantics of categorical attributes does match up well with the idea of a spatial **region**.

Categorical regions

- regions: contiguous bounded areas distinct from each other
 - using space to separate (proximity)
 - following expressiveness principle for categorical attributes
- use ordered attribute to order and align regions



List Alignment: One Key

- With a **single key**, separating into regions using that key yields one region per item.
- The regions are frequently arranged in a one-dimensional **list alignment**, either horizontal or vertical.
- The view itself covers a two-dimensional area: the aligned list of items stretches across one of the spatial dimensions, and the region in which the values are shown stretches across the other.

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Idiom: bar chart

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

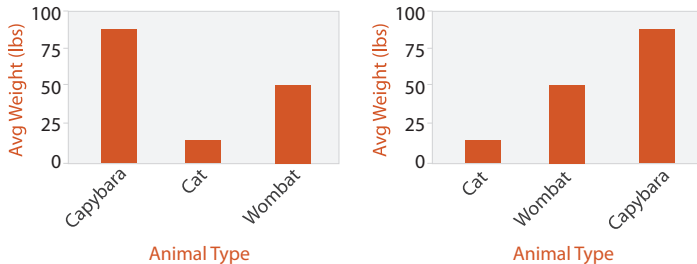
Space-Filling

Idiom	Bar Charts
What: Data	Table: one quantitative value attribute, one categorical key attribute.
How: Encode	Line marks, express value attribute with aligned vertical position, separate key attribute with horizontal position.
Why: Task	Lookup and compare values.
Scale	Key attribute: dozens to hundreds of levels.



Idiom: bar chart (cont.)

- Bar chart. The key attribute, species, separates the marks along the horizontal spatial axis. The value attribute, weight, expresses the value with aligned vertical spatial position and line marks. (*left*) Marks ordered alphabetically according to species name. (*right*) Marks ordered by the weight attribute used for bar heights.





Idiom: stacked bar chart

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Idiom	Stacked Bar Charts
What: Data	Multidimensional table: one quantitative value at tribute, two categorical key attributes.
How: Encode	Bar glyph with length-coded subcomponents of value attribute for each category of secondary key attribute. Separate bars by category of primary key attribute.
Why: Task	Part-to-whole relationship, lookup values, find trends.
Scale	Key attribute (main axis): dozens to hundreds of levels. Key attribute (stacked glyph axis): several to one dozen



Idiom: stacked bar chart (cont.)

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

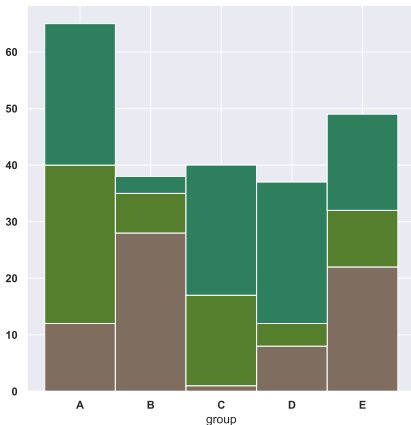
Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling





Idiom: streamgraph

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Idiom	Streamgraphs
What: Data	Multidimensional table: one quantitative value attribute (counts), one ordered key attribute (time), one categorical key attribute (artist).
What: Derived	One quantitative attribute (for layer ordering).
How: Encode	Use derived geometry showing artist layers across time, layer height encodes counts.
Scale	Key attributes (time, main axis): hundreds of time points. Key attributes (artists, short axis): dozens to hundreds

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

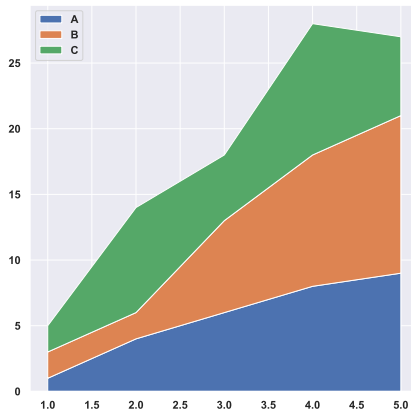
Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

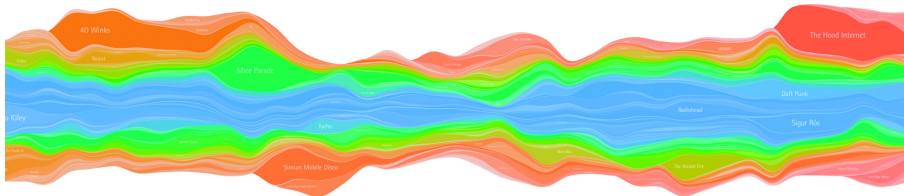
Idiom: streamgraph (cont.)





Idiom: streamgraph (cont.)

- Streamgraph of music listening history



Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

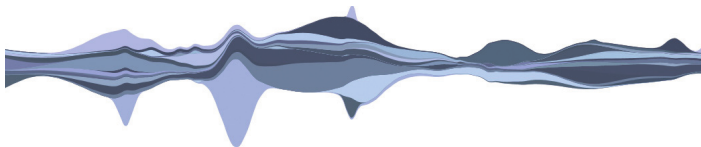
Dense

Space-Filling

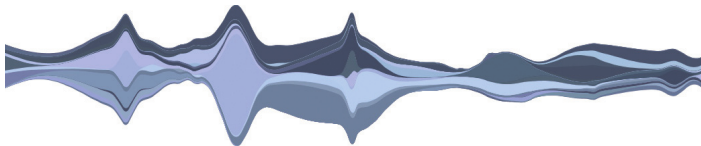


Idiom: streamgraph (cont.)

- Streamgraphs with layers ordered by different derived attributes. (a) Volatility of artist's popularity. (b) Onset time when artist's music of first gained attention.



(a)



(b)

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Idiom: line chart / dot chart

- The **dot chart** idiom is a visual encoding of one quantitative attribute using spatial position against one categorical attribute using point marks, rather than the line marks of a bar chart

Idiom	Dot Charts
What: Data	Table: one quantitative value attribute, one ordered key attribute.
How: Encode	Express value attribute with aligned vertical position and point marks. Separate/order into horizontal regions by key attribute.

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Idiom: line chart / dot chart (cont.)

- The idiom of **line charts** augments dot charts with line connection marks running between the points.

Idiom	Line Charts
What: Data	Table: one quantitative value attribute, one ordered key attribute.
How: Encode	Dot chart with connection marks between dots.
Why	Show trend.
Scale	Key attribute: hundreds of levels.

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

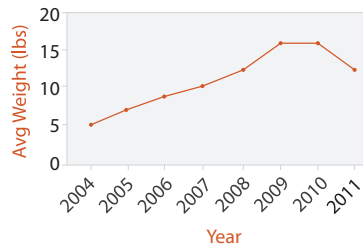
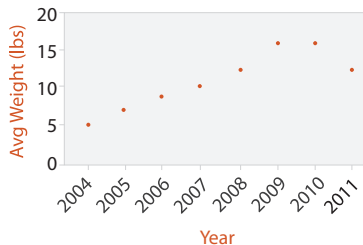
Dense

Space-Filling

Idiom: line chart / dot chart (cont.)



- Line charts versus dot charts. (*left*) Dot charts use a point mark to show the value for each item. (*right*) Line charts use point marks connected by lines between them.





Choosing bar vs. line charts

- Depends on type of key attributes
 - bar charts if categorical
 - line charts if ordered
- Do not use line charts for categorical key attributes
 - violates expressiveness principle
 - implication of trend so strong that it overrides semantics!

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Choosing bar vs. line charts (cont.)



- Bar charts and line charts both encode a single attribute. Bar charts encourage discrete comparisons, while line graphs encourage trend assessments. Line charts should not be used for categorical data, as in the upper right, because their implications are misleading → “The more male a person is, the taller he/she is”

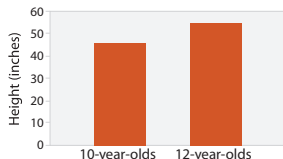
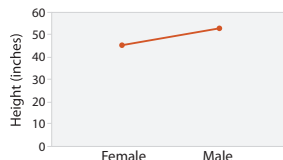
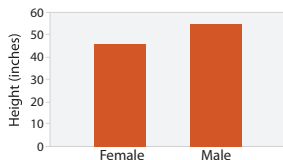




Chart axes

- Labelled axis is critical
- Avoid cropping y-axis: include 0 at bottom left or slope misleads
- Dual axes controversial
 - Acceptable if commensurate
 - Beware, very easy to mislead!

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Choosing line chart aspect ratios

When designing a line chart, an important question to consider is its **aspect ratio**: the ratio of width to height of the entire plot.

1. banking to 45° (1980s)

- Cleveland perceptual argument: most accurate angle judgement at 45°

2. multi scale banking to 45° (2006)

- frequency domain analysis to find ratios: FFT the data, convolve with Gaussian to smooth
- find interesting spikes/ranges in power spectrum: cull nearby regions if similar, ensure overview
- create trend curves (red) for each aspect ratio

3. arc length based aspect ratio (2011)

- minimize the arc length of curve while keeping the area of the plot constant



Choosing line chart aspect ratios (cont.)

- parametrization and scale invariant
- symmetry preserving
- robust & fast to compute

4. meta-points from this progression

- young field; prescriptive advice changes rapidly
- reasonable defaults required deep dive into perception meets math

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

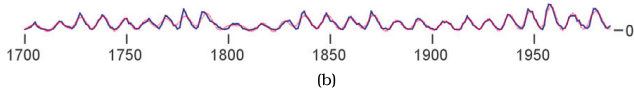
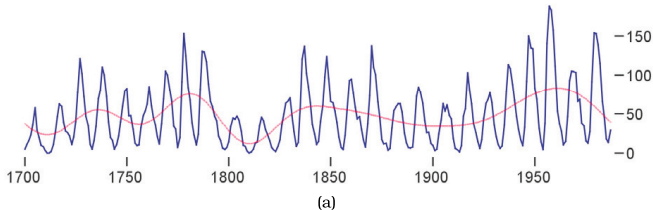
Dense

Space-Filling



Choosing line chart aspect ratios (cont.)

- Sunspot cycles. The multiscale banking to 45° idiom exploits our orientation resolution accuracy at the diagonal. (a) An aspect ratio close to 4 emphasizes low-frequency structure. (b) An aspect ratio close to 22 shows higher-frequency structure: cycle onset is mostly steeper than the decay





Idiom: connected scatterplots

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

- scatterplot with line connection marks
 - popular in journalism
 - horiz + vert axes: value attribs
 - line connection marks: temporal order
 - alternative to dual-axis charts
 - horiz: time
 - vert: two value attribs
- empirical study
 - engaging, but correlation unclear



Idiom: Indexed line charts

- data: 2 quantitative attributes
 - 1 key + 1 value
- derived data: new quantitative value attribute
 - index
 - plot instead of original value
- task: show change over time
 - principle: normalized, not absolute
- scalability: same as standard line chart

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Idiom: Gantt charts

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

- one key, two (related) values
- data
 - 1 categ attrib, 2 quant attribs
- mark: line
 - length: duration
- channels
 - horiz position: start time (+end from duration)
- task: emphasize temporal overlaps, start/end dependencies between items
- scalability: dozens of key levels, hundreds of value levels



Idiom: Slopegraphs

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

- two values
- data
 - 2 quant value attribs
- mark: point + line
 - line connecting mark between pts
- channels
 - 2 vertical pos: express attrib value
- task: emphasize changes in rank/value
- scalability: hundreds of value levels



Idiom: histogram

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

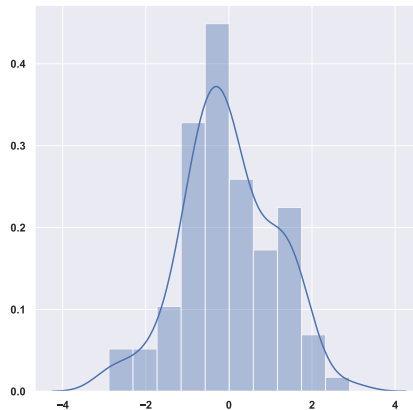
Spatial Layout
Density

Dense

Space-Filling

- static item aggregation
- task: find distribution
- data: table
- derived data
 - new table: keys are bins, values are counts
- scalability
 - depends on bin size, not original table size
- bin size crucial
 - pattern can change dramatically depending on discretization
 - opportunity for interaction: control bin size on the fly

Idiom: histogram (cont.)





Idiom: boxplot

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

- static item aggregation
- task: find distribution, find outliers
- data: table
- derived data
 - 5 quant attribs
 - median: central line
 - lower and upper quartile: boxes
 - lower upper fences: whiskers, values beyond which items are outliers
 - outliers beyond fence cutoffs explicitly shown
- scalability: constant, whatever size of original table



Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

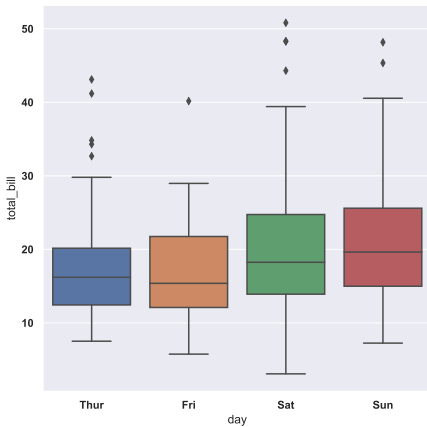
Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Idiom: boxplot (cont.)





Breaking conventions

- presentation vs. exploration
- engaging/evocative
 - inverted y axis
 - blood drips down on Poe

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Matrix Alignment: Two Keys

- Datasets with two keys are often arranged in a two-dimensional **matrix alignment** where one key is distributed along the rows and the other along the columns, so a rectangular cell in the matrix is the region for showing the item values

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Idiom: heatmap

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Idiom	Heatmaps
What: Data	Table: two categorical key attributes (genes, conditions), one quantitative value attribute (activity level for gene in condition).
How: Encode	2D matrix alignment of area marks, diverging color map.
Why: Task	Find clusters, outliers; summarize.
Scale	Items: one million. Categorical attribute levels: hundreds. Quantitative attribute levels: 3–11.



Idiom: heatmap (cont.)

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

**Matrix Alignment:
Two Keys**

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

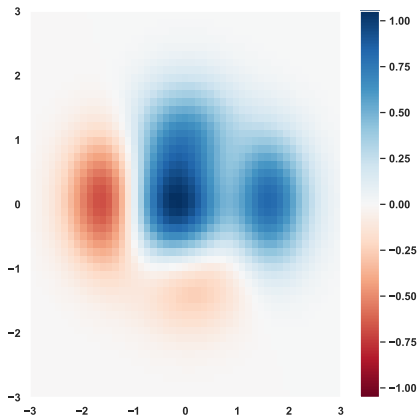
Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling





Idiom: cluster heatmap

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Idiom	Cluster Heatmaps
What: Derived How: Encode	Two cluster hierarchies for table rows and columns. Heatmap: 2D matrix alignment, ordered by both cluster hierarchies. Dendrogram: connection line marks for parent–child relationships in tree.



Idiom: cluster heatmap (cont.)

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

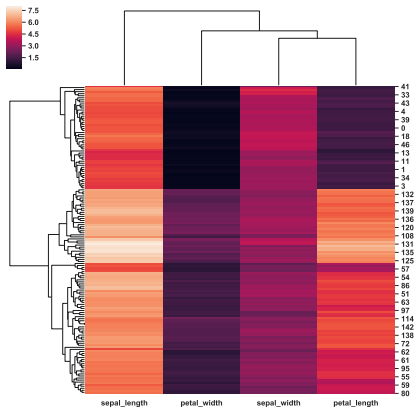
Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

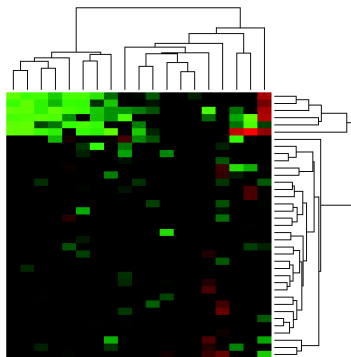
Space-Filling



Idiom: cluster heatmap (cont.)



- A heatmap provides a compact summary of a quantitative value attribute with 2D matrix alignment by two key attributes and small area marks colored with a diverging colormap. The cluster heatmap includes trees drawn on the periphery showing how the matrix is ordered according to the derived data of hierarchical clusterings on its rows and columns





Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Idiom: Scatterplot Matrix

- A **scatterplot matrix** (SPLOM) is a matrix where each cell contains an entire scatterplot chart. A SPLOM shows all possible pairwise combinations of attributes, with the original attributes as the rows and columns

Idiom	Scatterplot Matrix (SPLOM)
What: Data What: Derived How: Encode Why: Task Scale	Table. Ordered key attribute: list of original attributes. Scatterplots in 2D matrix alignment. Find correlation, trends, outliers. Attributes: one dozen. Items: dozens to hundreds.



Idiom: Scatterplot Matrix (cont.)

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

**Spatial Axis
Orientation**

Rectilinear Layouts

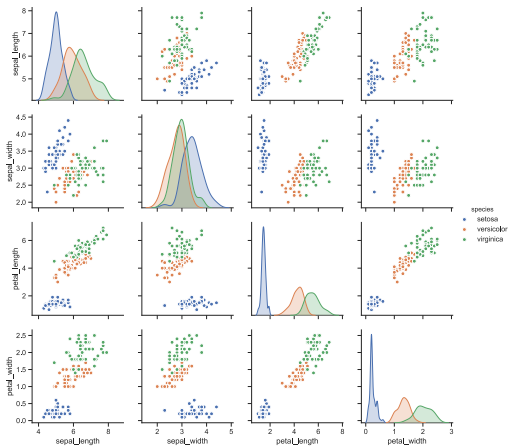
Parallel Layouts

Radial Layouts

**Spatial Layout
Density**

Dense

Space-Filling





Volumetric Grid: Three Keys

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

**Volumetric Grid: Three
Keys**

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

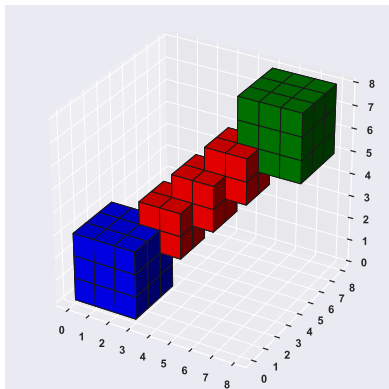
Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Recursive Subdivision: Multiple Keys





Spatial Axis Orientation

- Rectilinear Layouts
- Parallel Layouts
- Radial Layouts

Rectilinear Layouts



- In a **rectilinear** layout, regions or items are distributed along two perpendicular axes, horizontal and vertical spatial position, that range from minimum value on one side of the axis to a maximum value on the other side



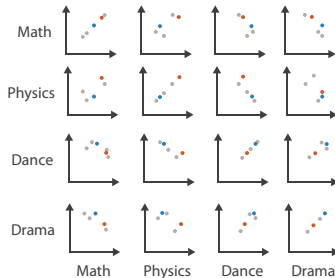
Parallel Layouts

- The idiom of **parallel coordinates** is an approach for visualizing many quantitative attributes at once using spatial position. As the name suggests, the axes are placed parallel to each other, rather than perpendicularly at right angles.

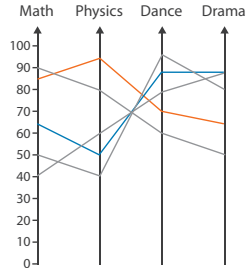
Table

Math	Physics	Dance	Drama
85	95	70	65
90	80	60	50
65	50	90	90
50	40	95	80
40	60	80	90

Scatterplot Matrix



Parallel Coordinates





Idiom: parallel coordinates

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

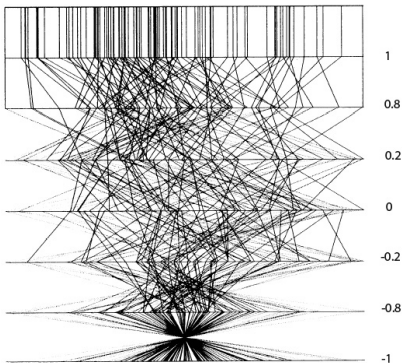
Space-Filling

Idiom	Parallel Coordinates
What: Data How: Encode Why: Tasks Scale	Table: many value attributes. Parallel layout: horizontal spatial position used to separate axes, vertical spatial position used to express value along each aligned axis with connection line marks as segments between them. Find trends, outliers, extremes, correlation. Attributes: dozens along secondary axis. Items: hundreds.



Idiom: parallel coordinates (cont.)

- Parallel coordinates were designed to show correlation between neighboring axes. At the top, parallel lines show perfect positive correlation. At the bottom, all of the lines cross over each other at a single spot in between the two axes, showing perfect negative correlation. In the middle, the mix of crossings shows uncorrelated data.



Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

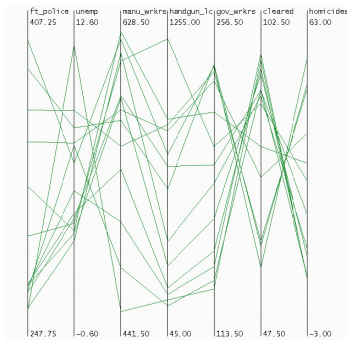
Dense

Space-Filling

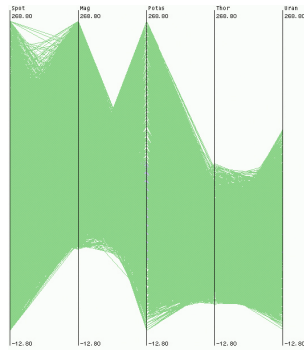


Idiom: parallel coordinates (cont.)

- Parallel coordinates scale to dozens of attributes and hundreds of items, but not to thousands of items. (a) Effective use with 13 items and 7 attributes. (b) Ineffective use with over 16,000 items and 5 attributes.



(a)



(b)



Task: Correlation

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

- scatterplot matrix
 - positive correlation → diagonal low-to-high
 - negative correlation → diagonal high-to-low
 - uncorrelated
- parallel coordinates
 - positive correlation → parallel line segments
 - negative correlation → all segments cross at halfway point
 - uncorrelated → scattered crossings



Orientation tradeoffs/limitations

- rectilinear: scalability with respect to the number of axes
 - 2 axes best
 - 3 often problematic: more later
 - 4+ impossible
- parallel: unfamiliarity, training time

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

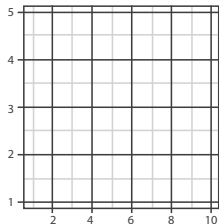
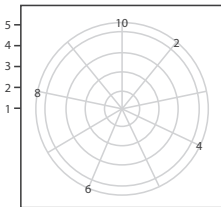
Dense

Space-Filling



Radial Layouts

- In a **radial** spatial layout, items are distributed around a circle using the angle channel in addition to one or more linear spatial channels, in contrast to the rectilinear layouts that use only two spatial channels.





Idioms: radial bar chart, star plot

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

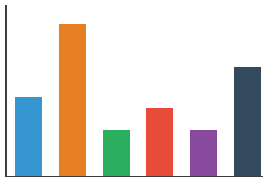
Idiom	Radial Bar Charts
What: Data	Table: one quantitative attribute, one categorical attribute.
How: Encode	Length coding of line marks; radial layout.

- radial bar chart: radial axes meet at central ring, line mark
- star plot: radial axes, meet at central point, line mark
- bar chart: rectilinear axes, aligned vertically
- accuracy: length unaligned with radial, less accurate than aligned with rectilinear



Idioms: radial bar chart, star plot (cont.)

- Radial versus rectilinear layouts. (*left*) Rectilinear bar chart. (*right*) Radial bar chart



Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Idioms: pie chart, polar area chart

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Idiom	Pie Charts
What: Data	Table: one quantitative attribute, one categorical attribute.
Why: Task	Part-whole relationship.
How: Encode	Area marks (wedges) with angle channel; radial layout.
Scale	One dozen categories.



Idioms: pie chart, polar area chart (cont.)

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Idiom	Polar Area Charts
What: Data	Table: one quantitative attribute, one categorical attribute.
Why: Task	Part-whole relationship.
How: Encode	Area marks (wedges) with length channel; radial layout.
Scale	One dozen categories.

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

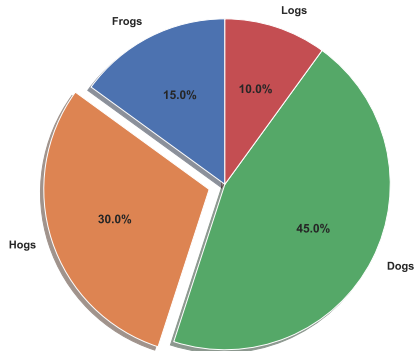
Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

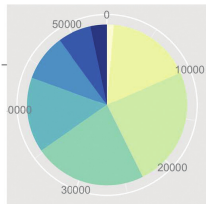
Idioms: pie chart, polar area chart (cont.)



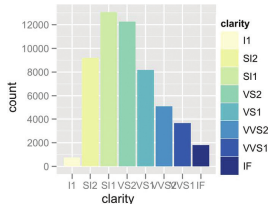


Idioms: pie chart, polar area chart (cont.)

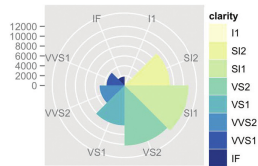
- Pie chart versus bar chart accuracy. (a) Pie charts require angle and area judgements. (b) Bar charts require only high-accuracy length judgements for individual items. (c) Polar area charts are a more direct equivalent of bar charts, where the length of each wedge varies like the length of each bar.



(a)



(b)



(c)

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts
Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Idioms: normalized stacked bar chart

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

Idiom	Normalized Stacked Bar Charts
What: Data	Multidimensional table: one quantitative value attribute, two categorical key attributes.
What: Derived	One quantitative value attribute (normalized version of original attribute).
Why: Task	Part-whole relationship.
How: Encode	Line marks with length channel; rectilinear layout.
Scale	One dozen categories for stacked attribute. Several dozen categories for axis attribute.



Idioms: normalized stacked bar chart (cont.)

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

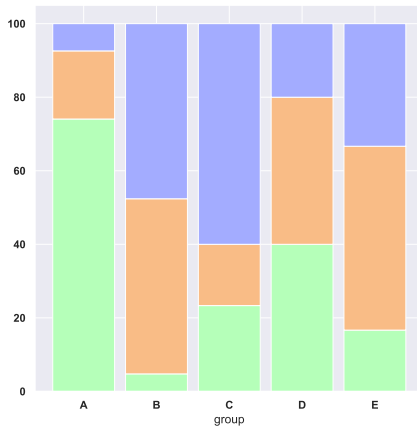
Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Matrix Alignment:
Two Keys

Recursive Subdivision: Multiple Keys

Rectilinear Layouts

Parallel Layouts

Radial Layouts

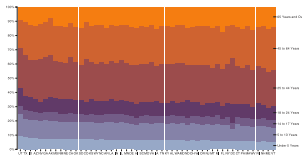
Spatial Layout Density

Dense

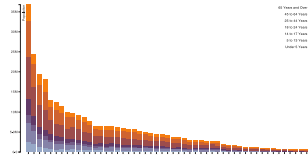
Space-Filling

-

(a)



(b)



(c)



Pie chart perception

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

- some empirical evidence that people respond to arc length
 - not angles
 - maybe also areas?...
- donut charts no worse than pie charts
- meta-points
 - redesign of paper figures in later blog post → violin plots good for analysis but too detailed for presentation
 - my advice: still dubious for pie/donut charts → sometimes ok if just 2 attribs



Rectilinear vs. Radial

- rectilinear good for linear vs nonlinear trends
- radial good for cyclic patterns
 - use case: finding periodicity
- radial orientation
 - perceptual limits
 - polar coordinate asymmetry
 - angles lower precision than lengths
 - frequently problematic
 - sometimes can be deliberately exploited! → for 2 attributes of very unequal importance

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling



Rectilinear vs. Radial (cont.)

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

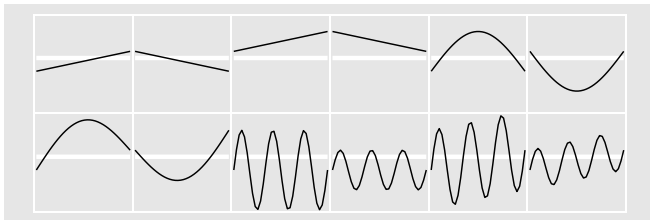
Parallel Layouts

Radial Layouts

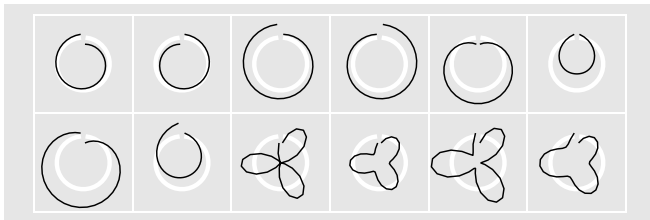
Spatial Layout
Density

Dense

Space-Filling



(a)



(b)



Spatial Layout Density

- Dense
- Space-Filling

Dense



- A **dense** layout uses small and densely packed marks to provide an overview of as many items as possible with very high information density.
- A maximally dense layout has point marks that are only a single pixel in size and line marks that are only a single pixel in width.



Idiom: Dense Software Overviews

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

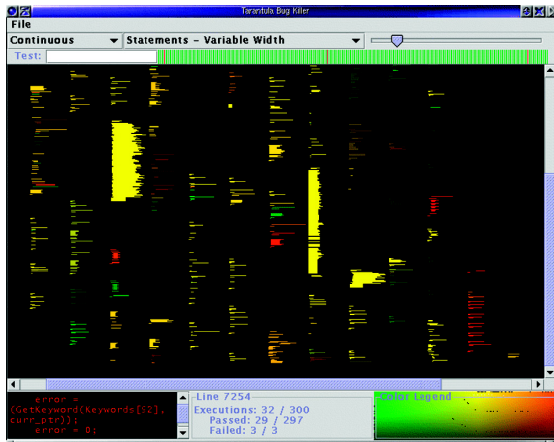
Space-Filling

Idiom	Dense Software Overviews
What: Data	Text with numbered lines (source code, test results log).
What: Derived	Two quantitative attributes (test execution results).
How: Encode	Dense layout. Spatial position and line length from text ordering. Color channels of hue and brightness. Locate faults, summarize results and coverage.
Why: Task	Lines of text: ten thousand.
Scale	Same encoding, same dataset, global overview with detail showing subset of data, different resolutions, linking with color.
(How: Facet)	
(How: Reduce)	Detail: filter to local neighborhood of selection



Idiom: Dense Software Overviews (cont.)

- Tarantula shows a dense overview of source code with lines color coded by execution status of a software test suite.





Space-Filling

- A **space-filling** layout has the property that it fills all available space in the view, as the name implies
- Space-filling layouts typically use area marks for items or containment marks for relationships, rather than line or connection marks, or point marks.

Express:
Quantitative
Values

Separate,
Order, and
Align:
Categorical
Regions

List Alignment: One
Key

Matrix Alignment:
Two Keys

Volumetric Grid: Three
Keys

Recursive Subdivision:
Multiple Keys

Spatial Axis
Orientation

Rectilinear Layouts

Parallel Layouts

Radial Layouts

Spatial Layout
Density

Dense

Space-Filling

References



Goodfellow, I., Bengio, Y., and Courville, A. (2016).

Deep learning.

MIT press.



Munzner, T. (2014).

Visualization analysis and design.

CRC press.



Russell, S. and Norvig, P. (2016).

Artificial intelligence: a modern approach.

Pearson Education Limited.



Ward, M. O., Grinstein, G., and Keim, D. (2015).

Interactive data visualization: foundations, techniques, and applications.

CRC Press.